TWO INVARIANTS OF AN EFFECT MONAD: CODENSITY GOVERNS FULLNESS, POLYNOMIALITY GOVERNS COMPOSITION

MACBETH

Date: 6 September 2026.

Abstract. Fix a monad M on **Set**. An *M-container* is an ordinary container (S, P) whose positions are read out in the Kleisli category of M; its extension is the endofunctor $[\![S, P]\!]_M(X) = \sum_{s \in S} \mathrm{Kl}(M)(P_s, X)$, the effectful reading of a data type whose observations may fail, branch, consult a context, or return a distribution. We prove three things about the extension functor $[\![-]\!]_M \colon M\text{-}\mathbf{Cont} \to [\mathbf{Set}, \mathbf{Set}]$. First, M-containers are exactly $\mathrm{Fam}(\mathrm{Kl}(M)^{\mathrm{op}})$ and the extension factors as $[\![-]\!]_M = [\![-]\!] \circ M$: an M-container functor is an ordinary polynomial functor post-composed with the effect monad. Second, $[\![-]\!]_M$ is *always faithful*, and it is *full* if and only if M is *codense* — the canonical map $M \Rightarrow \mathrm{Ran}_M M$ into its own codensity monad is an isomorphism — and each functor $\mathbf{Set}(A, M-)$ is connected; consequently probability distributions keep fullness while error, partiality, read-only context and nondeterminism destroy it. Third, the essential image of $[\![-]\!]_M$ is closed under composition *if* M is a *polynomial* functor — sufficiency for all M, with $p \triangleleft_M q = p \triangleleft \lceil M \rceil \triangleleft q$ — and the converse holds on a class delimited by an exact combinatorial condition, strictly larger than the *fully symmetric* world. Composition-closure forces M polynomial for every analytic monad with $M\emptyset = \emptyset$ (by *right-cancellation of plethysm* in the cycle-index ring, at any degree or nesting depth, covering free magmas and free operad monads), for every analytic monad of bounded *wreath depth* (by an imprimitivity-depth invariant subsuming the multiset and symmetric-power monads), and for the free-algebra monad of every unital non-flat operad with a *commutative binary composite* and $\Delta_{C_2} \notin \mathcal{S}$ (by a correlated block-swap no polynomial layer can produce; the condition is strictly weaker than full symmetry, covering the free commutative unital magma U and the A_4- and V_4-operads); but it *fails* for partially symmetric unital operads — the free algebra monad of a ternary operation with cyclic (C_3) slot-symmetry and an absorbed unit is non-polynomial yet has composition-closed image ($[\![p]\!]_M \circ [\![q]\!]_M \cong [\![p \triangleleft L]\!]_M$ with L the list monad). Commutativity of a binary composite is a *precondition* for the escape (it drives the swap), not by itself the line; given it, the dividing line is the finer condition $\Delta_{C_2} \in \mathcal{S}$ — whether the locked double-swap $\langle (0\,1)(2\,3) \rangle$ is the automorphism group of a single O-tree, characterised by (M1) a coexisting rigid and commutative binary composite, or (M2) a locked-swap derived operation. Thus *within the symmetric world of the applied effect monads — multiset, distribution, powerset — composition $\iff$ flatness of the species $\iff$ polynomiality*, with the C_3 operad marking the exact boundary. The affine-plus-commutative folklore criterion is false (D is affine and commutative but does not compose; Maybe is neither yet does). Fullness and composition are thus governed by two independent Kan-theoretic invariants of M that trade oppositely: probability is faithful but not composable, error is composable but not faithful.

Author: MacBeth (Kodamai). *Date:* 10 September 2026. *For review by:* Rick.

Corresponds to `scotmacbeth/work-in-progress` *commit* `4158115`.

Draft — work in progress; not yet neighbour-reviewed. This is the referee revision of commit `70becd9` (Rick, accept with minor revisions): it addresses the five flags and *sharpens* the necessity criterion. Necessity now holds not merely within the fully symmetric world but for the free-algebra monad of every unital non-flat operad with a commutative binary composite and $\Delta_{C_2} \notin \mathcal{S}$ (Theorem 5.22, §5.8), a class strictly containing full symmetry (e.g. the A_4- and V_4-operads) and hence resolving the corner U. The earlier "commutativity of the binary unit-composite is the dividing line" framing is corrected: commutativity is a precondition, and among operads that have a commutative binary composite the true dividing line is $\Delta_{C_2} \in \mathcal{S}$, characterised by (M1)$\vee$(M2) (Proposition 5.21). The C_3-ternary unital operad still *refutes* necessity outright (Theorem 5.24), and a false sharpness aside on plethysm cancellation has been withdrawn (§5.7). The one open question is the converse, $\Delta_{C_2} \in \mathcal{S} \Rightarrow$ closed (Problem 5.26).

1. Introduction

Containers, and what happens when the positions become effectful. A *container* (S, P) — a set S of shapes together with a set P_s of positions over each shape — presents a datatype through its extension, the endofunctor

$$[\![S, P]\!](X) \;=\; \sum_{s \in S} \mathbf{Set}(P_s, X) \;=\; \sum_{s \in S} X^{P_s},$$

which sends X to "a shape together with an X-labelling of its positions" [4, 6]. The extensions of containers are exactly the *polynomial functors*, and $[\![-]\!]$ is fully faithful: a natural transformation between two such functors is precisely a container morphism. This tight correspondence — a syntax of shapes and positions on one side, a class of functors on the other — is what makes containers a usable calculus for datatypes.

Now suppose the positions are not read out purely but *through an effect.* Reading a position might fail, or return several answers, or consult an environment, or produce a probability distribution over answers. All of these are captured by a monad M on $\mathbf{Set}$: an effectful read of a P_s-position landscape into X is a Kleisli map $P_s \to X$ in $\mathrm{Kl}(M)$, i.e. a function $P_s \to MX$. This yields the *M-container* extension

$$[\![S, P]\!]_M(X) \;=\; \sum_{s \in S} \mathrm{Kl}(M)(P_s, X) \;=\; \sum_{s \in S} (MX)^{P_s}.$$

The passage $\mathbf{Cont} \to M$-$\mathbf{Cont}$ that reinterprets a container's positions in $\mathrm{Kl}(M)$ is a change of base of exactly the kind one wants for modelling effectful data structures and effectful agents: an M-container is a device that, presented with a request, chooses a shape and then reads its positions with the ambient effect. When $M = \mathsf{Maybe}$, an "agent may decline" a position; when $M = \mathsf{D}$ (finite distributions), it answers probabilistically.

Two questions, and the temptation to conflate them. For the pure calculus, the extension functor is fully faithful and strong monoidal: it names functors faithfully, and those functors compose. Two properties, one embedding. When we move to M-containers, both properties become genuine questions:

(Q1) **Faithful naming.** Is $[\![-]\!]_M \colon M\text{-}\mathbf{Cont} \to [\mathbf{Set}, \mathbf{Set}]$ full and faithful? That is, does every natural transformation between M-container functors come from a (unique) M-container morphism?

(Q2) **Composition.** Is the essential image of $[\![-]\!]_M$ closed under functor composition? That is, if F and G are M-container functors, is $F \circ G$ one too, presented by some container built from theirs?

It is tempting to expect that these rise and fall together — that "M-containers behave well" is a single condition on M. The main discovery of this note is that (Q1) and (Q2) are controlled by *two different* invariants of M, and that across the standard effects they trade against each other.

Results. Throughout, $M = (M, \eta, \mu)$ is a monad on $\mathbf{Set}$, $\mathrm{Kl}(M)$ its Kleisli category, and $[\![-]\!] \colon \mathbf{Cont} \to [\mathbf{Set}, \mathbf{Set}]$ the ordinary (polynomial) extension.

The first theorem identifies the players and provides the lever for everything else.

Theorem 1.1 (Identification and factorisation; §3)**.** *The category of M-containers is* $\mathrm{Fam}(\mathrm{Kl}(M)^{\mathrm{op}})$*: an object is an ordinary container $(S, P \colon S \to \mathbf{Set})$, and a morphism $(S, P) \to (S', P')$ is a shape map $f \colon S \to S'$ together with, for each s, a Kleisli map $\rho_s \in \mathrm{Kl}(M)(P'_{fs}, P_s) = \mathbf{Set}(P'_{fs}, MP_s)$. Moreover, as functors $\mathbf{Set} \to \mathbf{Set}$ and naturally in (S, P),*

$$[\![S, P]\!]_M \;=\; [\![S, P]\!] \circ M.$$

So an M-container functor is an *ordinary polynomial functor pre-composed with the effect monad M*. Neil Ghani's "change of base" $\mathbf{Cont} \to M\text{-}\mathbf{Cont}$ is precomposition with the Kleisli inclusion $F \colon \mathbf{Set} \to \mathrm{Kl}(M)$, and every structural question about $[\![-]\!]_M$ becomes a question about the single trailing M.

The second theorem answers (Q1). Recall that the *codensity monad* of a functor $M \colon \mathbf{Set} \to \mathbf{Set}$ is the right Kan extension $\mathrm{Ran}_M M$; there is a canonical monad map $\iota_M \colon M \Rightarrow \mathrm{Ran}_M M$, and M is called *codense* when ι_M is an isomorphism (Definition 2.1). Call M *affine* if $M1 \cong 1$.

Theorem 1.2 (Faithfulness always; fullness = codensity; §4)**.** *The extension $[\![-]\!]_M \colon M\text{-}\mathbf{Cont} \to [\mathbf{Set}, \mathbf{Set}]$ is* always faithful*. It is* full *— hence fully faithful — if and only if*

(i) *each functor $\mathbf{Set}(A, M-)$ is connected, and*

(ii) *M is codense.*

If M is affine then (i) *holds; hence M* affine and codense $\Rightarrow$ fully faithful. *The finite distribution monad* D *is affine and codense, so* $[\![[\!]_M \mathsf{D}]-$ *is fully faithful* (*modulo the rigidity Lemma 4.4*)*; whereas* Maybe*, exceptions, the writer $E \times (-)$, the reader $(-)^R$ and nonempty powerset all fail codensity and so lose fullness.*

The correct invariant is the codensity monad, *not* preservation of coproducts. The natural first guess — "$[\![-]\!]_M$ is fully faithful iff M preserves coproducts" — is wrong for the target $[\mathbf{Set}, \mathbf{Set}]$: it is the criterion for a coarser, Kleisli-enriched target, and it gives the wrong verdict on D. The writer monad $E \times (-)$ preserves coproducts yet fails fullness for $|E| \geq 2$, a clean refutation (§4.3).

The third theorem answers (Q2). Call M *polynomial* if $M \cong \sum_{i \in I} (-)^{B_i}$ for some family $(B_i)_{i \in I}$; equivalently $M = [\![\lceil M \rceil]\!]$ for a container $\lceil M \rceil$.

Theorem 1.3 (Sufficiency of polynomiality; conditional converse; §5)**.** (Sufficiency, unconditional.) *If M is polynomial, then for all M-containers p, q the ordinary container*

$$p \triangleleft_M q \ := \ p \triangleleft \lceil M \rceil \triangleleft q$$

satisfies $[\![p \triangleleft_M q]\!]_M \cong [\![p]\!]_M \circ [\![q]\!]_M$, naturally in X. The operation $\triangleleft_M$ is associative and has a unit up to ordinary container isomorphism only when $M = \mathrm{Id}$*; so the essential image of $[\![-]\!]_M$ is closed under composition.* (Necessity, conditional.) *The converse — composition-closure $\Rightarrow$ M polynomial — holds for the following classes, but not in general. It holds if M grows super-polynomially (a cardinality law excludes powerset, nonempty powerset, distributions, ultrafilter, continuation); if M is analytic with $M\emptyset = \emptyset$ (right-cancellation of plethysm in the cycle-index ring, Theorem 5.18, at any degree or nesting depth, covering free magmas and free operad monads); if M is analytic of bounded* wreath depth *(an imprimitivity-depth invariant, Theorem 5.16, subsuming the multiset and symmetric-power monads); and if M is the free-algebra monad of a unital non-flat operad with a* commutative binary composite *and $\Delta_{C_2} \notin \mathcal{S}$ (conditions (i)+(ii) of Theorem 5.22) — a class containing every fully symmetric unital operad (hence the corner case U, formerly left open) and strictly more (e.g. the A_4- and V_4-operads).* But the converse fails *for partially symmetric unital operads: the free algebra monad of a single ternary operation with cyclic (C_3) slot-symmetry and an absorbed unit is analytic and non-polynomial, yet its essential image* is *closed under composition (Theorem 5.24). The exact dividing line is* not *the commutativity of the binary unit-composite $\mu(x, y) = \omega(x, y, e, \dots)$, as the two endpoints alone might suggest: commutativity is a* precondition *for the escape (a commutative composite is what supplies the swap), and given it the outcome is decided by the finer condition $\Delta_{C_2} \in \mathcal{S}$ — whether the locked double-swap $\langle (0\,1)(2\,3) \rangle$ is the automorphism group of some single O-tree,* *read off the operad by (M1) a coexisting rigid* and *commutative binary composite, or (M2) a derived operation carrying that symmetry (§5.8). Given a commutative binary composite and $\Delta_{C_2} \notin \mathcal{S}$ — as under full symmetry — $M \circ M$ realises a correlated block-swap that no polynomial layer $[\![r]\!] \circ M$ can manufacture, so necessity holds; the C_3 operad, whose only binary composite is rigid so that no swap can be built at all, sits outside this class and its image closes without polynomiality. Consequently* (*Theorem 5.27*) within the fully symmetric world — which contains all the applied effect monads: multiset, distribution, powerset — composition-closure $\iff$ flatness of the species $\iff$ polynomiality*, and the C_3 operad marks the precise boundary at which the symmetry hypothesis becomes load-bearing.*

Combining the two theorems gives the punchline.

Corollary 1.4 (Independence and trade-off)**.** *Fullness* (codensity) *and composability* (polynomiality) *are independent invariants of M. The writer monad $E \times (-)$ is polynomial, hence composes, but is not codense, hence not full. The distribution monad D is affine and codense, hence full, but is not polynomial, hence does not compose.* Probability keeps faithful naming but loses composition; error keeps composition but loses faithful naming.

Method. Everything runs off the factorisation $[\![-]\!]_M = [\![-]\!] \circ M$ of Theorem 1.1. For faithfulness, the Kleisli unit η is a section, so container data can be read back off the induced natural transformation. For fullness, precomposition $(-) \circ M$ has a right adjoint Ran_M, and every homset $\mathrm{Nat}(\mathbf{Set}(A, M-), H)$ is a value of a right Kan extension along M; the fullness obstruction is exactly the failure of the unit $M \Rightarrow \mathrm{Ran}_M M$ to be invertible. For composition, two applications of the factorisation put every composite $[\![p]\!]_M \circ [\![q]\!]_M$ in the form $G \circ M$ with $G = [\![p]\!] \circ M \circ [\![q]\!]$; this is again an M-extension precisely when G is polynomial, and then AAG strong monoidality delivers the splice $p \triangleleft \lceil M \rceil \triangleleft q$. Sufficiency is immediate. The converse is where the structure lives, and it is genuinely partial. The trailing M cannot be cancelled at the level of connected-limit preservation — Proposition 5.10 shows the single self-composite $M \circ M \cong [\![r]\!] \circ M$ carries no information about M beyond the retract reduction Lemma 5.9 — so the converse needs a *finer, faithful* invariant. The master lemma is that a *polynomial* outer layer is rigid on positions: every point-stabilizer of $([\![r]\!] \circ M)[m]$ is a *block-fixing product* of single-M-element stabilizers (Proposition 5.12, "a function is fixed iff fixed pointwise"). Three invariants then measure how far $M \circ M$ escapes this constraint: a cardinality count for super-polynomial growth; an imprimitivity-*wreath-depth* statistic h (Definition 5.14, see §5.6) for bounded-depth analytic monads; and right-cancellation of plethysm on the cycle-index series (Lemma 5.17) for analytic M with $M\emptyset = \emptyset$, reaching the unbounded-depth monads no stabilizer count can touch. The residual after these — analytic, $M\emptyset$ nonempty-finite, unbounded wreath depth — is where the fine structure of the operad becomes decisive, through an exact combinatorial condition on single-tree automorphism groups. Whenever the operad is unital and non-flat with a commutative binary composite and $\Delta_{C_2} \notin \mathcal{S}$ — no single O-tree has the locked double-swap $\langle (0\,1)(2\,3)\rangle$ as automorphism group — $M \circ M$ builds a *rigid* composite block from the unit and *swaps* two equal copies via that commutative composite, a correlated symmetry Δ_{C_2} that no block-fixing product contains, and necessity holds (Theorem 5.22); full symmetry is the special case in which this always applies. If instead the operad has a rigid binary composite — as for the C_3 operad, where $\omega(x, y, e)$ is rigid and non-commutative — the escape is unavailable, and an "only three identical" rigidity (Lemma 5.25) shows every stabilizer of its self-composite is already a single-tree automorphism group, so the image closes and necessity can fail (Theorem 5.24).

Why it matters. Beyond the pleasing tension of Corollary 1.4, the result sharpens the compositional picture of effectful agents. Modelling "an agent may decline" as a Maybe-container, one keeps composition (agents chain into agents) at the price of faithfulness (distinct declining strategies can induce the same functor). Modelling "an agent answers probabilistically" as a D-container, one keeps faithful naming (strategies are recoverable from behaviour) but loses strict composition (a chain of probabilistic agents is not itself presented by a single probabilistic container — the marginal-to-joint gap). The invariant that decides each is now explicit and computable: codensity for the first question, polynomiality for the second.

Provenance and honesty. Theorems 1.1 and 1.2 (faithfulness, the codensity criterion, the affine lemma, the examples) are proved. Theorem 1.3 is proved in its sufficiency direction unconditionally, and in its necessity direction for each of the four classes named: the cardinality law disposes of the super-polynomial monads; plethysm right-cancellation (Lemma 5.17, Theorem 5.18) disposes of every analytic monad with $M\emptyset = \emptyset$, at unbounded degree and depth; the wreath-depth invariant (Definition 5.14, Theorem 5.16) disposes of every bounded-wreath-depth analytic monad, subsuming the multiset and symmetric-power monads; and the correlated-swap witness (Theorem 5.22) disposes of every unital non-flat operad with a commutative binary composite and $\Delta_{C_2} \notin \mathcal{S}$ (conditions (i)+(ii)), a class that strictly contains the fully symmetric operads and hence resolves the corner case U. This sharpens the earlier "full symmetry" criterion: the load-bearing conditions are a commutative binary composite and $\Delta_{C_2} \notin \mathcal{S}$, proved for all arities via the characterisation Proposition 5.21, with the supporting witnesses ($O_{\mu\nu}$, the pentagon D_5, the A_4- and V_4-operads) checked at $m = 4$. The failure of necessity for the C_3 operad (Theorem 5.24) is likewise proved: the isomorphism $M \circ M \cong L \circ M$ is established by matching molecule multisets arity-by-arity (the "only three identical" rigidity, Lemma 5.25); the small cases $m \leq 4$ are confirmatory computation, not the proof. The cancellation and block-fixing engines are elementary and computationally verified. (We have also removed a false aside from an earlier draft — that a vanishing linear coefficient breaks plethysm right-cancellation; it does not, and Lemma 5.17's hypothesis is sufficient but not necessary, §5.7.) Theorems 5.18, 5.16, 5.22 and 5.24 invoke two standard facts of combinatorial-species theory — Joyal's full faithfulness of the species-to-analytic-functor embedding, and the identity of species substitution with plethysm of cycle-index series (Joyal; Bergeron–Labelle–Leroux; Gambino–Kock for the polynomial- $\iff$ -flat criterion) — which we cite as background rather than re-derive. What remains genuinely open is the *converse* of Theorem 5.22 — whether $\Delta_{C_2} \in \mathcal{S}$ forces the essential image closed, which would upgrade the necessity criterion to a biconditional (Problem 5.26); it is verified for the all-binary inhabitant $O_{\mu\nu}$ at $m \leq 4$ and open in general. Full faithfulness for D rests on a standard rigidity fact, $\mathrm{Nat}(\mathsf{D}, \mathsf{D}) = \{\mathrm{id}\}$, which we isolate as Lemma 4.4 and assume rather than reprove. These boundaries are marked at each occurrence.

Outline. Section 2 fixes notation: monads and Kleisli categories, the container extension and its composition monoidal structure, families over an opposite category, and right Kan extensions and codensity monads. Section 3 proves Theorem 1.1 and the always-faithfulness. Section 4 proves the codensity criterion (Theorem 1.2), works the examples, and explains why coproduct preservation is the wrong criterion. Section 5 proves Theorem 1.3: sufficiency and non-unitality (§5.2); the cardinality law (§5.3); the block-fixing meta-theorem and stabilizer invariant (§5.5); the wreath-depth invariant and Theorem S′ (§5.6); the plethysm right-cancellation closing the analytic converse for $M\emptyset = \emptyset$ (§5.7); and the unital corner (§5.8), where a correlated-swap witness and an exact criterion $\Delta_{C_2} \notin \mathcal{S}$ give necessity for a class strictly larger than the fully symmetric operads, while the C_3 operad refutes it, pinning $\Delta_{C_2} \in \mathcal{S}$ — not the mere commutativity of a binary composite — as the exact dividing line. Section 6 draws the trade-off and lists open problems.

2. Preliminaries

We work over **Set**. All the constructions below have counterparts over a general base along the lines of $\mathrm{Fam}(\mathcal{C}^{\mathrm{op}})$ [3], but the codensity and polynomiality phenomena are already visible, and cleanest, over **Set**.

2.1. **Monads and Kleisli categories.** A monad $M = (M, \eta, \mu)$ on **Set** has unit $\eta\colon \mathrm{Id} \Rightarrow M$ and multiplication $\mu\colon MM \Rightarrow M$. Its *Kleisli category* $\mathrm{Kl}(M)$ has sets as objects, $\mathrm{Kl}(M)(A, B) = \mathbf{Set}(A, MB)$, identity η_A, and composition $g \bullet f = \mu \circ Mg \circ f$. The free functor $F\colon \mathbf{Set} \to \mathrm{Kl}(M)$ is the identity on objects and sends $g\colon A \to B$ to $\eta_B \circ g$; it is faithful and bijective on objects, is a left adjoint (so preserves colimits), and is *not* full for nontrivial M. We call M *affine* if $M1 \cong 1$, and *commutative* if its two canonical strengths $MA \times MB \to M(A \times B)$ agree.

2.2. **Containers and the polynomial extension.** A *container* is a pair (S, P) with S a set and $P\colon S \to \mathbf{Set}$ (write P_s for the fibre). The category **Cont** has as morphisms $(S, P) \to (S', P')$ the pairs (f, φ) of a shape map $f\colon S \to S'$ and a reindexing $\varphi_s\colon P'_{fs} \to P_s$; equivalently $\mathbf{Cont} = \mathrm{Fam}(\mathbf{Set}^{\mathrm{op}})$ (see below). The *extension* is

$$[\![S, P]\!](X) = \sum_{s\in S} \mathbf{Set}(P_s, X), \qquad [\![S, P]\!](g) = \textstyle\sum_s (g \circ -).$$

The functor $[\![-]\!]\colon \mathbf{Cont} \to [\mathbf{Set}, \mathbf{Set}]$ is fully faithful, with essential image the *polynomial functors* $\sum_{i\in I}(-)^{B_i}$ [4, §1],[6]. We say M is *polynomial* when M lies in this image, and write $\lceil M\rceil$ for a container with $[\![\lceil M\rceil]\!] = M$.

Cont carries the *composition* monoidal structure $(\mathbf{Cont}, \triangleleft, \mathsf{y})$, where $\mathsf{y} = (1, 1 \mapsto 1)$ is the identity container ($[\![\mathsf{y}]\!] = \mathrm{Id}$) and $\triangleleft$ is defined so that $[\![-]\!]$ is strong monoidal:

$$[\![c \triangleleft d]\!] \;\cong\; [\![c]\!] \circ [\![d]\!], \qquad [\![\mathsf{y}]\!] = \mathrm{Id}, \tag{1}$$

naturally, and $\triangleleft$ is associative up to coherent isomorphism [4, Thm. 1.12],[6]. Directed containers — containers whose $\triangleleft$-structure is a comonoid — are the categories internal to this calculus [1].

2.3. **Families over an opposite category.** For a category $\mathcal{C}$, the category $\mathrm{Fam}(\mathcal{C})$ has objects $(S, X\colon S \to \mathrm{Ob}\,\mathcal{C})$ and morphisms $(S, X) \to (S', X')$ given by a function $f\colon S \to S'$ and a family of $\mathcal{C}$-maps $X_s \to X'_{fs}$. Thus $\mathrm{Fam}(\mathcal{C}^{\mathrm{op}})$ has the same objects but morphisms whose component maps run $X'_{fs} \to X_s$ in $\mathcal{C}$. In particular $\mathbf{Cont} = \mathrm{Fam}(\mathbf{Set}^{\mathrm{op}})$.

2.4. **Right Kan extensions and the codensity monad.** Precomposition $M^* = (-) \circ M\colon [\mathbf{Set}, \mathbf{Set}] \to [\mathbf{Set}, \mathbf{Set}]$ has a right adjoint, the right Kan extension $\mathrm{Ran}_M\colon [\mathbf{Set}, \mathbf{Set}] \to [\mathbf{Set}, \mathbf{Set}]$ along M. The defining adjunction gives, for $H\colon \mathbf{Set} \to \mathbf{Set}$ and any representable $\mathbf{Set}(A, -)$,

$$\begin{aligned} \mathrm{Nat}\big(\mathbf{Set}(A, M-),\, H\big) &= \mathrm{Nat}\big(\mathbf{Set}(A, -) \circ M,\, H\big) \\ &= \mathrm{Nat}\big(\mathbf{Set}(A, -),\, \mathrm{Ran}_M H\big) = (\mathrm{Ran}_M H)(A), \end{aligned} \tag{2}$$

using the Yoneda lemma in the last step. Applying this with $H = M$ produces the *codensity monad* of M.

Definition 2.1. The *codensity monad* of M is $\Phi := \mathrm{Ran}_M M$; by (2), $\Phi(A) = \mathrm{Nat}(\mathbf{Set}(A, M-), M)$. The unit η of the adjunction $M^* \dashv \mathrm{Ran}_M$, evaluated at M, is a natural monad map $\iota_M\colon M \Rightarrow \Phi$, explicitly $\iota_{M,A}(m) = [\,k \mapsto \mu_A(Mk\,(m))\,]$. We call M *codense* when ι_M is an isomorphism.

Definition 2.2. A functor $G\colon \mathbf{Set} \to \mathbf{Set}$ is *connected* if $\mathrm{Nat}(G, -)$ preserves coproducts: $\mathrm{Nat}(G, \sum_j H_j) = \sum_j \mathrm{Nat}(G, H_j)$ for every family (H_j). Equivalently, every natural transformation out of G into a coproduct factors through a single summand.

3. The identification, and always-faithfulness

Definition 3.1. An *M-container* is a container (S,P); its *M-extension* is

$$[\![S,P]\!]_M(X) = \sum_{s\in S} \mathrm{Kl}(M)(P_s,X) = \sum_{s\in S} \mathbf{Set}(P_s,MX),$$

with action $[\![S,P]\!]_M(g) = \sum_s (Mg\circ -)$ on a map $g\colon X\to X'$. A morphism $(S,P)\to(S',P')$ of M-containers is a shape map $f\colon S\to S'$ together with, for each s, a Kleisli map $\rho_s \in \mathrm{Kl}(M)(P'_{fs},P_s) = \mathbf{Set}(P'_{fs},MP_s)$; composition uses $\bullet$ in $\mathrm{Kl}(M)$.

Proof of Theorem 1.1. (a) M-$\mathbf{Cont} = \mathrm{Fam}(\mathrm{Kl}(M)^{\mathrm{op}})$. Objects of $\mathrm{Fam}(\mathrm{Kl}(M)^{\mathrm{op}})$ are pairs $(S, P\colon S\to \mathrm{Ob}\,\mathrm{Kl}(M))$, and $\mathrm{Ob}\,\mathrm{Kl}(M) = \mathbf{Set}$, so they are exactly containers. A morphism $(S,P)\to(S',P')$ is $f\colon S\to S'$ together with, per s, a $\mathrm{Kl}(M)^{\mathrm{op}}$-map $P_s\to P'_{fs}$, i.e. a $\mathrm{Kl}(M)$-map $P'_{fs}\to P_s$, i.e. a function $P'_{fs}\to MP_s$ — exactly Definition 3.1. Identities match (Fam's identity uses id in $\mathrm{Kl}(M)^{\mathrm{op}}$, i.e. η, the M-container identity), and composition matches (Fam composes shapes in $\mathbf{Set}$ and positions by $\bullet$ in the opposite order). The two categories are equal on objects, hom-sets, identities and composition.

(b) $[\![S,P]\!]_M = [\![S,P]\!]\circ M$. On objects, $[\![S,P]\!](MX) = \sum_s \mathbf{Set}(P_s,MX) = [\![S,P]\!]_M(X)$; on $g\colon X\to X'$ both sides act by $\sum_s(-\circ Mg)$. For naturality in (S,P), a morphism (f,ρ) induces $\alpha_X(s,k) = (fs,\ k\bullet\rho_s) = (fs,\ \mu_X\circ Mk\circ\rho_s)$, which is precisely $[\![-]\!]\circ M$ applied to the underlying shape/position data. □

Part (b) is the lever: *an M-container functor is a polynomial functor with a trailing M.*

Proposition 3.2 (Always faithful)**.** *$[\![-]\!]_M\colon M$-$\mathbf{Cont}\to[\mathbf{Set},\mathbf{Set}]$ is faithful.*

Proof. Suppose (f,ρ) and $(\tilde f,\tilde\rho)$ induce the same natural transformation α. Evaluate at $X = P_s$ on the element $(s,\eta_{P_s}) \in [\![S,P]\!]_M(P_s)$: $\alpha_{P_s}(s,\eta_{P_s}) = (fs,\ \eta_{P_s}\bullet\rho_s) = (fs,\rho_s)$, since η is the Kleisli identity. Equality of α therefore forces $fs = \tilde f s$ and $\rho_s = \tilde\rho_s$ for every s. Hence $(f,\rho) = (\tilde f,\tilde\rho)$. □

Remark 3.3. Faithfulness uses only that η is a section of the Kleisli identity; it holds for every monad. This already separates M-containers from linear (vector-space) containers, whose extension is not faithful because the analogue of η is not a section.

4. Fullness is codensity

Fix M-containers (S,P) and (S',P'). By Theorem 1.1(b), $[\![S,P]\!]_M = p\circ M$ and $[\![S',P']\!]_M = p'\circ M$ with $p = [\![S,P]\!]$, $p' = [\![S',P']\!]$. Since $p = \sum_s \mathbf{Set}(P_s,-)$ is a coproduct of representables,

$$\mathrm{Nat}(p\circ M,\ p'\circ M) = \prod_s \mathrm{Nat}\Big(\mathbf{Set}(P_s,M-),\ \sum_{s'} \mathbf{Set}(P'_{s'},M-)\Big), \tag{3}$$

while the M-container hom-set is $\prod_s \sum_{s'} \mathbf{Set}(P'_{s'}, MP_s)$. Comparing, fullness holds for all pairs if and only if, for every set A (playing the role of P_s) and every family $(P'_{s'})$, the map

$$(\star)\qquad \begin{aligned} \sum_{s'} \mathbf{Set}(P'_{s'}, MA) \;&\longrightarrow\; \mathrm{Nat}\Big(\mathbf{Set}(A, M-),\ \sum_{s'} \mathbf{Set}(P'_{s'}, M-)\Big), \\ (s', \rho) \;&\mapsto\; [\, k \mapsto (s', k \bullet \rho)\,], \end{aligned}$$

is a bijection.

4.1. **The Kan engine and the split.** By (2) with A, every natural transformation out of $\mathbf{Set}(A, M-)$ is a value of a right Kan extension along M. Two features of $\mathbf{Set}$ let us split $(\star)$ cleanly. First, a single summand is a product of copies of M: $\mathbf{Set}(B, M-) = \prod_{b\in B} M$, so $\mathrm{Nat}(\mathbf{Set}(A, M-), \mathbf{Set}(B, M-)) = \mathrm{Nat}(\mathbf{Set}(A, M-), M)^B = \Phi(A)^B$ by Definition 2.1. Second, the coproduct over s' is handled exactly when $\mathbf{Set}(A, M-)$ is connected. This yields:

Proof of Theorem 1.2. Faithfulness is Proposition 3.2. For fullness we show $(\star)$ is a bijection for all $A, (P'_{s'})$ iff (i) and (ii) hold.

(ii) handles a single summand. With one summand $B = P'_{s'}$, $(\star)$ becomes $\mathbf{Set}(B, MA) \to \mathrm{Nat}(\mathbf{Set}(A, M-), \mathbf{Set}(B, M-)) = \Phi(A)^B$, which is $\big(MA \to \Phi(A)\big)^B$; and the map is $\iota_{M,A}$ in each coordinate. So the single-summand case for all B is a bijection iff $\iota_{M,A}\colon MA \to \Phi(A)$ is a bijection for all A — that is, iff M is codense.

(i) handles the coproduct. Given codensity, the general $(\star)$ is a bijection iff the canonical comparison, with $G_{s'} = \mathbf{Set}(P'_{s'}, M-)$,

$$\textstyle\sum_{s'} \mathrm{Nat}(\mathbf{Set}(A, M-), G_{s'}) \;\longrightarrow\; \mathrm{Nat}(\mathbf{Set}(A, M-), \sum_{s'} G_{s'})$$

is a bijection — that is, iff $\mathbf{Set}(A, M-)$ is connected.

Hence $(\star)$ for all data $\iff$ (i)$\wedge$(ii), proving the criterion. ∎

Lemma 4.1 (Affine $\Rightarrow$ connected)**.** *If M is affine then each $\mathbf{Set}(A, M-)$ is connected; hence M affine and codense $\Rightarrow$ $[\![-]\!]_M$ is fully faithful.*

Proof. Since $M1 \cong 1$, $\mathbf{Set}(A, M1) = (M1)^A \cong 1$: the functor $\mathbf{Set}(A, M-)$ has a single global point. Let $\alpha\colon \mathbf{Set}(A, M-) \Rightarrow \sum_{s'} G_{s'}$. The unique point of $\mathbf{Set}(A, M1)$ maps to a single summand s'_0 of $\sum_{s'} G_{s'}(1)$. For any X, the terminal map $!\colon X \to 1$ gives a commuting square; since $\sum_{s'} G_{s'}(!)$ preserves summands and the image of the global point lies in s'_0, naturality forces α_X to land in the s'_0-summand for every X. Thus α factors through s'_0, which is connectedness. ∎

4.2. **The examples.** For a polynomial monad $M = \sum_{i\in I} (-)^{B_i}$ the whole comparison is computable by (1): $\mathbf{Set}(A, M-)$ is again polynomial, with shapes $\varphi\colon A \to I$ and positions $\sum_a B_{\varphi a}$, so that

$$|MA| = \sum_i |A|^{|B_i|}, \qquad |\Phi(A)| = \prod_{\varphi\colon A\to I} \sum_{i\in I} \Big(\textstyle\sum_a |B_{\varphi a}|\Big)^{|B_i|}.$$

Codensity is the equality $|MA| = |\Phi(A)|$ for all A. Exact computation gives:

M	$M1$	$\lvert MA\rvert$ vs $\lvert\Phi(A)\rvert$ $(A = 0, 1, 2, 3)$	codense?	full?
Id	1	$0, 1, 2, 3 \quad = \quad 0, 1, 2, 3$	yes	**yes**
Maybe $(X+1)$	2	$1, 2, 3, 4$ vs $1, 2, \mathbf{12}, \mathbf{864}$	no	no
exception $X+2$	3	$2, 3, 4, 5$ vs $2, \mathbf{12}, \mathbf{5184}, \ldots$	no	no
writer $2\times X$	2	$0, 2, 4, 6$ vs $0, \mathbf{4}, \mathbf{256}, \ldots$	no	no
reader X^2	1	$0, 1, 4, 9$ vs $0, \mathbf{4}, \mathbf{16}, \mathbf{36}$	no	no

Beyond a low threshold $|\Phi(A)| > |MA|$ strictly in every nontrivial case: the surplus natural transformations are "case-analysis" operations — inspecting $\perp$, or duplicating a reader argument — that are natural for pure maps but not Kleisli-natural.

Example 4.2 (Reader: affine but not codense)**.** The reader monad $(-)^R$ is affine ($M1 = 1$, so (i) holds by Lemma 4.1) yet fails (ii): $\Phi(A) = (RA)^R \neq A^R = MA$ for $|R| \geq 2$. Affineness alone does not give fullness; codensity is a genuinely separate necessary condition.

Example 4.3 (Distributions: affine and codense)**.** Let D be the finitely-supported distribution monad. It is affine ($\mathsf{D}1 \cong 1$). We claim it is codense: $\Phi(A) = \mathrm{Nat}(\mathsf{D}^A, \mathsf{D}) = \mathsf{D}A$, i.e. the natural A-ary operations on D are exactly the affine (convex) combinations. Given $\alpha\colon \mathsf{D}^A \Rightarrow \mathsf{D}$ and a tuple $(\mu_a)_a \in (\mathsf{D}X)^A$, put $K = \bigsqcup_a \mathrm{supp}(\mu_a)$, let $\nu_a \in \mathsf{D}K$ be the block-a copy of μ_a, and $p\colon K \to X$ the projection; then $(\mu_a) = \mathsf{D}^A(p)((\nu_a))$, so $\alpha_X((\mu_a)) = \mathsf{D}p\,(\alpha_K((\nu_a)))$. Collapsing the blocks by $c\colon K \to A$ gives $\mathsf{D}c\,(\alpha_K((\nu_a))) = \alpha_A((\delta_a)) =: \omega \in \mathsf{D}A$. Reducing one block at a time (collapse all but one block to a point) reduces to classifying $\mathrm{Nat}(\mathsf{D}, \mathsf{D}(-\sqcup F))$ for finite F: the F-marginal is constant (it factors through $\mathsf{D}1 = 1$) and the remaining marginal is a natural sub-probability-valued endofunctor of D, hence a scalar $\lambda \cdot \sigma$. Thus $\alpha_K((\nu_a)) = \sum_a \omega_a \nu_a$ and $\alpha_X((\mu_a)) = \sum_a \omega_a \mu_a$. So D is codense, and by Lemma 4.1 $[\![[\!]_M \mathsf{D}]-$ *is fully faithful* — modulo the base rigidity Lemma 4.4.

Lemma 4.4 (Rigidity of D; assumed)**.** *The only natural endomorphism of* D *is the identity, and every natural sub-probability-valued endofunctor of* D *is a scalar multiple of the identity:* $\mathrm{Nat}(\mathsf{D}, \mathsf{D}) = \{\mathrm{id}\}$.

This is a standard rigidity fact for the affine functor D; we assume it rather than reprove it, and we mark the full faithfulness of $[\![[\!]_M \mathsf{D}]-$ as conditional on it. The reduction in Example 4.3 is unconditional.

4.3. **Why coproduct preservation is the wrong criterion.** A natural first guess is that $[\![-]\!]_M$ is fully faithful iff M preserves coproducts. This is false for the target $[\mathbf{Set}, \mathbf{Set}]$, for a structural reason. Factor

$$[\![-]\!]_M \;=\; R \circ [\![-]\!]^{\mathrm{Kl}}, \qquad R = (- \circ F),$$

where $[\![-]\!]^{\mathrm{Kl}}\colon \mathrm{Fam}(\mathrm{Kl}(M)^{\mathrm{op}}) \to [\mathrm{Kl}(M), \mathrm{Kl}(M)]$ is the free-coproduct-completion extension (always fully faithful) and R restricts along the free functor $F\colon \mathbf{Set} \to \mathrm{Kl}(M)$. "$M$ preserves coproducts" is the fullness criterion for the coarser target $[\mathrm{Kl}(M), \mathrm{Kl}(M)]$ — the connected-unit condition for $[\![-]\!]^{\mathrm{Kl}}$ — and $[\![-]\!]^{\mathrm{Kl}}$ is already fully faithful. All the genuine content is in R, i.e. in the gap between pure-naturality and Kleisli-naturality, and that gap is measured by the codensity monad, not by coproduct preservation.

The verdicts differ concretely. Coproduct preservation fails for D, so the wrong criterion predicts D fails fullness; the correct criterion (affine and codense) says D *keeps* fullness. Conversely the writer monad $E \times (-)$ *preserves* coproducts, so the wrong criterion predicts fullness, yet $\Phi(1) = |E|^{|E|} \neq |E|$ for $|E| \geq 2$, so $E \times (-)$ is not codense and $[\![-]\!]_M$ is not full. The writer monad is thus a clean refutation of the coproduct criterion.

Remark 4.5 (Corrected slogan)**.** Adding probabilistic positions preserves the on-the-nose faithful naming of the container extension; adding error, partiality, read-only context or nondeterminism destroys it. The dividing invariant is codensity of the effect monad, computed by the right Kan extension $\mathrm{Ran}_M M$.

5. Composition is polynomiality

We now turn to (Q2). Throughout, $[\![-]\!]\colon \mathbf{Cont} \to [\mathbf{Set}, \mathbf{Set}]$ is the strong monoidal extension of (1).

5.1. The structural identity.

Lemma 5.1. *For all M-containers p, q, as functors $\mathbf{Set} \to \mathbf{Set}$ and naturally in X,*

$$[\![p]\!]_M \circ [\![q]\!]_M \;=\; G \circ M, \qquad G := [\![p]\!] \circ M \circ [\![q]\!].$$

Proof. Two applications of Theorem 1.1(b) and associativity of $\circ$:

$$([\![p]\!]_M \circ [\![q]\!]_M)(X) = [\![p]\!]_M\big([\![q]\!](MX)\big) = [\![p]\!]\big(M([\![q]\!](MX))\big) = ([\![p]\!] \circ M \circ [\![q]\!] \circ M)(X).$$

□

So every composite of two M-extensions has a trailing M; it is again an M-extension exactly when the inner G is polynomial.

5.2. Sufficiency.

Proposition 5.2 (Sufficiency)**.** *If M is polynomial with container $\lceil M \rceil$, then $p \triangleleft_M q := p \triangleleft \lceil M \rceil \triangleleft q$ satisfies $[\![p \triangleleft_M q]\!]_M \cong [\![p]\!]_M \circ [\![q]\!]_M$ naturally in X. The operation $\triangleleft_M$ on* **Cont** *is associative up to the $\triangleleft$-associator.*

Proof. With $M = [\![\lceil M \rceil]\!]$, (1) applied twice gives

$$G = [\![p]\!] \circ M \circ [\![q]\!] = [\![p]\!] \circ [\![\lceil M \rceil]\!] \circ [\![q]\!] \cong [\![p \triangleleft \lceil M \rceil \triangleleft q]\!] = [\![r]\!], \qquad r := p \triangleleft \lceil M \rceil \triangleleft q,$$

naturally, the two bracketings agreeing by associativity of $\triangleleft$. Whiskering by M on the right and invoking Lemma 5.1,

$$[\![p]\!]_M \circ [\![q]\!]_M = G \circ M \cong [\![r]\!] \circ M = [\![r]\!]_M,$$

naturally in X. Associativity of $\triangleleft_M$ is associativity of $\triangleleft$: $(p \triangleleft_M q) \triangleleft_M s = p \triangleleft \lceil M \rceil \triangleleft q \triangleleft \lceil M \rceil \triangleleft s = p \triangleleft_M (q \triangleleft_M s)$. $\square$

Proposition 5.3 (Non-unitality)**.** *The operation $\triangleleft_M$ has a two-sided unit up to ordinary container isomorphism if and only if $M = \mathrm{Id}$.*

Proof. If $M = \mathrm{Id}$ then $\lceil M \rceil = \mathsf{y}$ and $p \triangleleft_M q = p \triangleleft q$, with unit y. Conversely, let e be a two-sided unit up to **Cont**-isomorphism. Taking $q = \mathsf{y}$ in the left unit law, $e \triangleleft_M \mathsf{y} = e \triangleleft \lceil M \rceil \cong \mathsf{y}$; applying the fully faithful $[\![-]\!]$ and (1) gives $[\![e]\!] \circ M \cong \mathrm{Id}$. Write $[\![e]\!] = \sum_{s \in S_e} \mathbf{Set}(E_s, -)$, so $\sum_s \mathbf{Set}(E_s, M-) \cong \mathrm{Id}$. Evaluate:

- At $X = 1$: $\sum_s |M1|^{|E_s|} = 1$. Each summand is ≥ 1 (as $|M1| \geq 1$ by η_1), so there is a unique shape s_0 with $|M1|^{|E_{s_0}|} = 1$; and $E_{s_0} = \emptyset$ is impossible ($\mathbf{Set}(\emptyset, M-) = \mathrm{const}_1$ gives value $1 \neq 2$ at $X = 2$). Hence $|M1| = 1$ and $\mathbf{Set}(E_{s_0}, M-) \cong \mathrm{Id}$.
- At $X = 2$: $|M2|^{|E_{s_0}|} = 2$. If $|E_{s_0}| \geq 2$ this is 1 or ≥ 4, never 2; so $|E_{s_0}| = 1$ and $|M2| = 2$.

Thus $E_{s_0} \cong 1$ and $\mathbf{Set}(1, M-) = M \cong \mathrm{Id}$. $\square$

Remark 5.4. The clean statement is a unit up to *ordinary container* isomorphism, obtained by pushing the unit law through the fully faithful $[\![-]\!]$. A unit up to the coarser M-**Cont** (Kleisli) isomorphism is a more delicate question we do not address. For applications the operative fact is: $(\mathbf{Cont}, \triangleleft_M)$ is semigroupal, and monoidal only at $M = \mathrm{Id}$.

Remark 5.5 (Scope on morphisms)**.** Proposition 5.2 defines $\triangleleft_M$ on objects and makes $[\![-]\!]_M$ strong semigroupal on the subcategory of *pure* morphisms (those of the form $\eta \circ g$). Extending $\triangleleft_M$ to a bifunctor on all of M-**Cont** so that $[\![-]\!]_M$ is strong semigroupal requires transporting horizontal composites back along $[\![-]\!]_M$, which is free exactly when $[\![-]\!]_M$ is full — i.e. when M is codense (Theorem 1.2). Thus the object-level closure holds for every polynomial M, and upgrades to the full Kleisli category precisely when M is additionally codense. This Kleisli-bifunctoriality is the exact meeting point of Theorems 1.2 and 1.3; we leave it open.

5.3. **Necessity: the cardinality growth law.** Sufficiency looks like half of a biconditional, but the converse is subtler. We prove a cardinality obstruction that already decides all the affine and commutative candidates, then isolate the one remaining categorical gap.

Proposition 5.6 (Cardinality growth law)**.** *Suppose the essential image $\{[\![r]\!] \circ M : r \in \mathbf{Cont}\}$ is closed under $\circ$. Then for every container q there is a fixed nonnegative-integer polynomial R_q with, writing $m(n) = |M(n)|$ and*

Q the cardinality polynomial of $[\![q]\!]$,

$$m\big(Q(m(n))\big) = R_q\big(m(n)\big) \qquad \text{for all finite } n.$$

In particular, at $q = \mathsf{y}$ (so $Q = \mathrm{Id}$), $m(m(n)) = R(m(n))$: the function m agrees on $\mathrm{im}(m)$ with a fixed nonnegative-integer polynomial.

Proof. The identity container y has $[\![\mathsf{y}]\!]_M = M$, and every $[\![q]\!]_M = [\![q]\!] \circ M$ lies in the image. Closure applied to $p = \mathsf{y}$ and general q gives, via Lemma 5.1, $M \circ [\![q]\!] \circ M = [\![\mathsf{y}]\!]_M \circ [\![q]\!]_M \cong [\![r_q]\!] \circ M$ for some container r_q. Taking cardinalities on a finite set of size n yields $m(Q(m(n))) = R_q(m(n))$ with R_q the (fixed) cardinality polynomial of r_q. The case $q = \mathsf{y}$ is $Q = \mathrm{Id}$. □

Corollary 5.7 (The affine/commutative folklore is false)**.** *Nonempty powerset $\mathcal{P}^+$, powerset $\mathcal{P}$ and the distribution monad D all fail composition, already at $p = q = \mathsf{y}$; while* Maybe *and the writer $E \times (-)$ compose. In particular "composes $\iff$ commutative and affine" is false in both directions.*

Proof. For $\mathcal{P}^+$, $m(n) = 2^n - 1$ and $m(m(n)) = 2^{2^n-1} - 1$ grows doubly exponentially in n, hence super-polynomially in $v = m(n)$; no fixed polynomial R satisfies $R(2^n - 1) = 2^{2^n-1} - 1$ for all n, so Proposition 5.6 fails at $q = \mathsf{y}$. The same argument applies to $\mathcal{P}$ ($m(n) = 2^n$). For D, the finite support map $\mathsf{D} \to \mathcal{P}^+$ is a monad morphism onto the affine-commutative finite structure, reproducing the $\mathcal{P}^+$ growth on cardinalities, so D fails likewise. On the other side, $\mathsf{Maybe} = X+1$ and $E \times (-)$ are polynomial, so they compose by Proposition 5.2. Both $\mathcal{P}^+$ and D are affine and commutative yet fail; Maybe and the writer are non-affine yet compose. □

5.4. **The categorical converse.** Proposition 5.6 is necessary but does not by itself pin M as a functor: cardinalities cannot separate M from a polynomial functor with the same finite counts. The clean categorical converse we want is

Lemma 5.8 (the converse statement)**.** *Consider the implication: if the essential image of $[\![-]\!]_M$ is closed under $\circ$ — equivalently, $M \circ N \circ M$ is of the form $[\![r]\!] \circ M$ for every polynomial N — then M is polynomial.*

We establish Lemma 5.8 for four classes of M (super-polynomial growth; analytic with $M\emptyset = \emptyset$; bounded wreath depth; free commutative unital magmas), and we exhibit a monad for which it *fails* (the C_3 operad, §5.8). The frame is classical: over **Set**, *M is polynomial $\iff$ M preserves connected limits* (wide pullbacks together with cofiltered limits) $\iff$ M is familially representable (Carboni–Johnstone; [4, §1], [6]). Two features cut the problem down.

First, M is a monad, hence a retract of its square.

Lemma 5.9 (Retract reduction)**.** *For a monad M on* **Set***: M is polynomial $\iff$ $M \circ M$ preserves connected limits.*

Proof. (⇒) Polynomial functors are closed under composition [4], so $M \circ M$ is polynomial and preserves connected limits. (⇐) The right-unit law $\mu \circ \eta M = 1_M$ exhibits M as a *split retract* of $M \circ M$ (section ηM, retraction μ). Limit-preservation passes to retracts — if the comparison map for $M \circ M$ is invertible then so is the one for its retract M, by a diagram chase using naturality of comparison maps in the functor variable. Hence M preserves connected limits, so M is polynomial. $\square$

Second, the trailing M cannot be cancelled, and this is now an exact statement rather than a worry. Closure at $p = q = \mathsf{y}$ yields only the single self-composite identity $M \circ M \cong [\![r]\!] \circ M$, and that identity is categorically inert.

Proposition 5.10 (The single instance is inert). *If $M \circ M \cong P \circ M$ with P polynomial and nonconstant, then for every connected-limit diagram D, $M \circ M$ preserves $D \iff M$ preserves D. Hence the hypothesis $M \circ M \cong [\![r]\!] \circ M$ carries no information about connected-limit preservation of M beyond Lemma 5.9.*

Proof. Naturally isomorphic functors preserve the same limits, so it suffices to treat $P \circ M$. The comparison map for the composite factors as $\varphi_{PM} = \big(P(\lim MD) \cong \lim(P \circ MD)\big) \circ P(\varphi_M)$, the left isomorphism holding because P, being polynomial, preserves the connected limit $\lim(MD)$ [4, §1]. A nonconstant polynomial $P = \sum_i (-)^{B_i}$ (some $B_i \neq \emptyset$) reflects isomorphisms, so φ_{PM} is invertible $\iff$ φ_M is. Thus $M \circ M$ preserves $D \iff M$ does. $\square$

This is the precise form of the trailing-M obstruction: precomposition $(-) \circ M$ is not conservative, and the self-composite relates M only to itself, so any proof of Lemma 5.8 must use either the *full* family $\{M \circ N \circ M \cong [\![r_N]\!] \circ M : N \text{ polynomial}\}$ or a genuinely different invariant. We supply the latter — a permutation-group invariant read off the automorphisms of each structure — and trace exactly how far it reaches, and where it stops.

5.5. **The stabilizer invariant and block-fixing rigidity.** A natural isomorphism of **Set**-endofunctors is $\mathrm{Aut}(X)$-equivariant at each X (naturality at automorphisms $\sigma \colon X \to X$), so it preserves point-stabilizers.

Lemma 5.11 (Stabilizer invariant). *If $\Theta \colon F \cong G$ then $\mathrm{Stab}_{FX}(x) = \mathrm{Stab}_{GX}(\Theta_X x) \subseteq \mathrm{Aut}(X)$ for all X, x. Hence the family of subgroups $\mathcal{S}(F) := \{\mathrm{Stab}_{FX}(x)\}$, up to conjugacy, is an isomorphism invariant of F. Read on the* each-label-once *components below, it records one subgroup per structure and so is defined even when a component is infinite.*

The master fact is that a *polynomial* outer layer is rigid on positions. Call a structure of an analytic functor *each-label-once* on $[m]$ if its content is $[m]$ with every label used exactly once (the multilinear part of the species component).

Proposition 5.12 (Block-fixing rigidity of polynomial precomposition)**.** *Let* $[\![r]\!] = \sum_s \mathbf{Set}(B_s, -)$ *be polynomial and* M *any endofunctor. On each each-label-once component, every point-stabilizer of* $([\![r]\!] \circ M)[m]$ *is a* block-fixing product $\prod_{\beta \in \pi} \mathrm{Stab}_M(u_\beta)$ *of single-*M*-element stabilizers, over the partition* π *of* $[m]$ *into the disjoint supports of the parts. In particular no two* M*-parts are ever permuted.*

Proof. An element of $([\![r]\!] \circ M)[m]$ is $(s, \varphi\colon B_s \to M[\,\cdot\,])$ with total content $[m]$. For $\sigma \in S_m$, $([\![r]\!] \circ M)(\sigma)(s, \varphi) = (s, M\sigma \circ \varphi)$, which equals (s, φ) iff $M\sigma(\varphi(b)) = \varphi(b)$ for *every* $b \in B_s$: a function is fixed iff fixed pointwise, there being no quotient on the positions B_s. So $\mathrm{Stab}(s, \varphi) = \bigcap_b \mathrm{Stab}(\varphi(b))$. Each-label-once forces the parts $\varphi(b)$ of nonempty content to have *disjoint* supports β partitioning $[m]$ (a shared label would be used twice), and for disjoint blocks $\bigcap_\beta [\,\mathrm{Aut} \text{ on } \beta \times \mathrm{Sym}(\text{rest})\,] = \prod_\beta \mathrm{Aut}$ on β. □

The each-label-once reading is essential and non-circular: on the *raw* set $([\![r]\!] \circ M)(X)$, where labels may repeat, the stabilizer is an intersection over parts with possibly overlapping supports — a strictly larger class of subgroups. It is exactly the disjointness on the multilinear component that collapses the intersection to a block-fixing product, and it is on that component that Lemma 5.11 is compared. The first application recovers the universal analytic monad that evades the cardinality law.

Theorem 5.13 (The multiset monad does not compose)**.** *Let* $\mathbb{M}$ *be the finite-multiset (free commutative monoid) monad,* $\mathbb{M}X = \{$*finite multisets on* $X\}$*. Then* $\mathbb{M} \circ \mathbb{M} \not\cong P \circ \mathbb{M}$ *for every polynomial* P*; in particular* $\mathbb{M}$ *does not have composition-closed image.*

Proof. A multiset $\nu \in \mathbb{M}X$ is a finitely-supported $\nu\colon X \to \mathbb{N}$, with $\mathrm{Stab}_{\mathbb{M}X}(\nu) = \prod_k \mathrm{Sym}(\nu^{-1}(k))$ a *Young subgroup* — it permutes points only *within* the level-blocks of ν. Intersections of Young subgroups are Young, so by Proposition 5.12 every stabilizer of $(P \circ \mathbb{M})[m]$ is a Young subgroup. But for X containing four distinct points a, b, c, d, the element $m = \{\{a, b\}, \{c, d\}\} \in \mathbb{M}\mathbb{M}X$ (content each-once) has

$$\mathrm{Stab}_{\mathbb{M}\mathbb{M}X}(m) = \big(\mathrm{Sym}\{a, b\} \times \mathrm{Sym}\{c, d\}\big) \rtimes \langle (a\,c)(b\,d) \rangle = S_2 \wr S_2 = D_4, \quad |D_4| = 8,$$

the within-block symmetries *together with the block-swap* $(a\,c)(b\,d)$. D_4 contains $(a\,c)(b\,d)$ but not the within-a-block transposition $(a\,c)$ (which would send m to $\{\{c, b\}, \{a, d\}\} \neq m$), so D_4 is not a Young subgroup: a Young subgroup containing $(a\,c)(b\,d)$ would place a, c in one block and hence also contain $(a\,c)$. (Concretely the Young subgroup orders in S_4 are $\{1, 2, 4, 6, 24\}$, and 8 is not among them.) Thus $D_4 \in \mathcal{S}(\mathbb{M}\mathbb{M})$ but $\mathcal{S}(P\mathbb{M})$ consists of Young subgroups only, and Lemma 5.11 gives $\mathbb{M}\mathbb{M} \not\cong P\mathbb{M}$. □

The block-swap is an irreducibly second-order symmetry that a free (polynomial) outer layer cannot manufacture: a single inner $\mathbb{M}$ supplies only within-block (Young) symmetry, whereas a non-free *outer* layer supplies between-equal-block (wreath) symmetry. In the species calculus this is exactly the distinction between the labelled products in $B \bullet A$ (direct-product stabilizers) and the wreath constituents of the self-substitution $A \bullet A$.

The multiset witness is a single wreath $S_2 \wr S_2$ against Young subgroups; it reaches only *bounded* nesting. For monads whose structures nest to unbounded depth — free magmas, free operad-algebra monads, with automorphism towers $S_2 \wr \cdots \wr S_2$ — no fixed stabilizer suffices, and the right invariant is the *depth* of the wreath tower.

5.6. **The wreath-depth invariant.** Recall the analytic/species dictionary (Joyal [5]; Bergeron–Labelle–Leroux [2]; Gambino–Kock [4]). An *analytic* functor is $\tilde{A}(X) = \sum_{n \geq 0} A[n] \times_{S_n} X^n$ for a *species* A (each $A[n]$ a finite S_n-set); the embedding $A \mapsto \tilde{A}$ is fully faithful, so $\tilde{A} \cong \tilde{B}$ iff $A[n] \cong B[n]$ as S_n-sets for all n (Joyal). Its molecular decomposition is $A = \coprod_l X^{d_l}/H_l$ with $H_l \leq S_{d_l}$ the automorphism group of a representative A-structure of arity d_l; $\tilde{A}$ is *polynomial* iff A is *flat* (every $H_l = 1$, equivalently every S_n-action free), by familial representability [4, §1]. Self-composition is species substitution, $\tilde{A} \circ \tilde{A} = \widetilde{A \bullet A}$. By Proposition 5.12 every stabilizer of $(\llbracket r \rrbracket \circ \tilde{A})[m]$ is a block-fixing product of single-tree stabilizers, each a $\mathrm{Sym}(\beta)$-conjugate of some H_l; we separate $\tilde{A} \circ \tilde{A}$ from $\llbracket r \rrbracket \circ \tilde{A}$ by a conjugacy invariant of subgroups that block-fixing products cannot inflate.

Definition 5.14 (Imprimitivity depth)**.** For a transitive $T \leq \mathrm{Sym}(\Omega)$, a *block system* is a T-invariant partition; let $h_t(T)$ be the maximal length of a strictly refining chain of block systems from $\{\Omega\}$ to the singletons (equivalently, of subgroups from a point-stabilizer up to T). For general $K \leq \mathrm{Sym}(\Omega)$, set $h(K) = \max\{h_t(K^O) : O \text{ an orbit}, \ |O| \geq 2\}$, and 0 if K is trivial, where K^O is the transitive group K induces on O. The *wreath depth* of a species is $\mathfrak{h}(A) := \sup_l h(H_l)$; A has *bounded wreath depth* iff $\mathfrak{h}(A) < \infty$.

Both h_t and h are conjugacy invariants (conjugation carries block systems to block systems). Sample values: $h_t(S_n) = 1$ (primitive groups have no proper nontrivial block system), $h_t(S_a \wr S_b) = 2$, $h_t(S_2 \wr S_2 \wr S_2) = 3$.

Lemma 5.15 (Wreath superadditivity; product depth)**.** *For transitive T_1, T_2 in the product action, $h_t(T_1 \wr T_2) \geq h_t(T_1) + h_t(T_2)$. For a block-fixing product $\prod_\beta G_\beta$ on pairwise disjoint supports, $h(\prod_\beta G_\beta) = \max_\beta h(G_\beta)$.*

Proof. For the wreath: the partition $\{\{j\} \times \Omega_1 : j \in \Omega_2\}$ is invariant with quotient action T_2; a longest chain of T_2 lifts to $h_t(T_2)$ strict steps above it, and a longest chain of T_1 applied uniformly in every block gives $h_t(T_1)$ strict steps below it, concatenating to a chain of length $\geq h_t(T_1) + h_t(T_2)$. For the product: the orbits of $\prod_\beta G_\beta$ are exactly the orbits of the individual G_β (each factor fixes the other supports), so $h(\prod_\beta G_\beta) = \max_\beta h(G_\beta)$. $\square$

Theorem 5.16 (Wreath-depth converse)**.** *Let $M = \tilde{A}$ be an analytic monad of bounded wreath depth $W = \mathfrak{h}(A) < \infty$. If A is non-flat then $\tilde{A} \circ \tilde{A} \not\cong$*

$[\![r]\!] \circ \tilde{A}$ for every polynomial $[\![r]\!]$; hence closure of the essential image forces M polynomial. This subsumes the bounded-degree case (degree $\leq d$ implies $\mathfrak{h} \leq d-1 < \infty$) and, at $W = 1$, reproves Theorem 5.13 and the entire symmetric-power class ($\mathfrak{h}(\mathbb{M}) = 1$ since each $H_n = S_n$ is primitive).

Proof. *(Bound.)* By Proposition 5.12 and Lemma 5.15, every stabilizer of $([\![r]\!] \circ \tilde{A})[m]$ is a block-fixing product of $\mathrm{Sym}(\beta)$-conjugates of constituent groups H_l, so has depth $\leq \max_\beta h(H_{l_\beta}) \leq \mathfrak{h}(A) = W$. *(Witness.)* Since A is non-flat, some constituent c acts nontrivially on an orbit ($h_t(H_c^{O_c}) \geq 1$); since $W < \infty$, the sup is attained by a constituent b with an orbit $h_t(H_b^{O_b}) = W$. Fill every slot of c with the same each-label-once b-structure on disjoint labels: the resulting $w \in \widetilde{A \bullet A}$ has $\mathrm{Stab}(w) = H_b \wr H_c$, and restricting to the invariant sub-orbit $O_c \times O_b$ gives the transitive sub-wreath $H_b^{O_b} \wr H_c^{O_c}$, of depth $\geq h_t(H_c^{O_c}) + h_t(H_b^{O_b}) \geq 1 + W$ by Lemma 5.15. So $\tilde{A} \circ \tilde{A}$ realises depth $\geq W + 1 > W$, and Lemma 5.11 gives $\tilde{A} \circ \tilde{A} \not\cong [\![r]\!] \circ \tilde{A}$. □

The invariant h is blind precisely when $\mathfrak{h}(A) = \infty$: then A already contains arbitrarily deep wreath towers, $\tilde{A} \circ \tilde{A}$ produces no *new* depth, and both sides have infinite depth. That residual splits by $a_0 = |M\emptyset|$: for $a_0 = 0$ an algebraic cancellation reaches every depth (§5.7); for $a_0 > 0$ the answer turns on symmetry (§5.8).

5.7. **Cycle-index cancellation: the case $M\emptyset = \emptyset$.** For unbounded wreath depth the invariant h cannot separate, but when $M\emptyset = \emptyset$ an algebraic cancellation on cycle indices does, at any depth. The *cycle-index series* of a species A is

$$Z_A = \sum_{n \geq 0} \frac{1}{n!} \sum_{\sigma \in S_n} \mathrm{fix}(A[\sigma])\, p_\sigma \in R := \mathbb{Q}[[p_1, p_2, \dots]], \qquad p_\sigma = \textstyle\prod_k p_k^{c_k(\sigma)}.$$

The facts we need, beyond Joyal full faithfulness **(J1)** above. **(F′)** $\tilde{A}$ is polynomial iff $Z_A \in \mathbb{Q}[[p_1]]$ (flatness in cycle-index form: a non-identity permutation fixing a structure contributes a p_λ with $\lambda \neq (1^n)$). **(C)** if $B[0] = \emptyset$ then species substitution is plethysm on cycle indices, $Z_{A \bullet B} = Z_A \circ Z_B$, where $\circ$ is the unique $\mathbb{Q}$-algebra endomorphism of R with $p_k \circ H = H[p_i \mapsto p_{ki}]$. Writing $a_0 = |A[0]| = |M\emptyset|$ and $a_1 = |A[1]|$, a monad has $a_1 \geq 1$ (the unit $\mathrm{Id} \Rightarrow \tilde{A}$ requires an S_1-fixed point of $A[1]$).

The engine is elementary.

Lemma 5.17 (Plethysm right-cancellation)**.** *Let $H \in R$ have zero constant term and nonzero linear coefficient, $H = a_1 p_1 + (\deg \geq 2)$ with $a_1 \neq 0$. Then $(-) \circ H \colon R \to R$ is injective: $F \circ H = G \circ H \Rightarrow F = G$.*

Proof. $(-) \circ H$ is a $\mathbb{Q}$-algebra endomorphism, so it suffices that $D \circ H = 0 \Rightarrow D = 0$. Since H has bottom degree 1 with $H^{(1)} = a_1 p_1$, plethysm gives $p_\lambda \circ H = a_1^{\ell(\lambda)} p_\lambda + (\deg > |\lambda|)$. If $D \neq 0$, let m be its bottom degree;

then $(D\circ H)^{(m)}=\sum_{|\lambda|=m} c_\lambda\, a_1^{\ell(\lambda)} p_\lambda$, and linear independence of $\{p_\lambda\}_{|\lambda|=m}$ together with $a_1\neq 0$ forces every $c_\lambda=0$, contradicting minimality. $\square$

The linear-coefficient hypothesis is what the bottom-degree bookkeeping above uses, and it holds in our application: $H=Z_A$ has $a_1\geq 1$, forced by the monad unit. It is worth flagging that the hypothesis is *sufficient but not necessary.* An elementary algebraic-independence argument (the images $p_k\circ H$ are algebraically independent because each carries a top-index variable the others lack) shows $(-)\circ H$ is injective for *every* non-constant polynomial H; right-cancellation of $\circ$ collapses only when H is a constant, where $F\circ H=F(H,H,\dots)$ flattens everything. In particular the example $H=p_2+\frac{1}{2}p_1^2$, with vanishing linear coefficient, still yields an *injective* $(-)\circ H$ (its kernel is zero in every degree, checked exactly) — so a zero linear coefficient does not by itself break cancellation. Only the sufficient direction, at $a_1\geq 1$, is used below.

Theorem 5.18 (Cancellation converse; $M\emptyset=\emptyset$)**.** *Let $M=\tilde{A}$ be an analytic monad with $A[0]=\emptyset$. If the essential image of $[\![-]\!]_M$ is closed under composition then M is polynomial — with no bound on degree or nesting depth.*

Proof. Closure at $p=q=\mathsf{y}$ gives $M\circ M\cong [\![r]\!]\circ M$ for a polynomial $[\![r]\!]$. As $M\circ M$ is analytic it is finitary, so $[\![r]\!]\circ\tilde{A}$ is finitary; and $\tilde{A}X\supseteq A[1]\times X$ (using $a_1\geq 1$) is unbounded in $|X|$, so an infinite exponent B_s in $[\![r]\!]=\sum_s(-)^{B_s}$ would make $(-)^{B_s}\circ\tilde{A}$ fail to preserve a filtered colimit — forcing all B_s finite. Thus $[\![r]\!]=\tilde{B}$ for a *flat* species B (fact (F′)). Evaluating the isomorphism at $\emptyset$ and using $\tilde{A}\emptyset=A[0]=\emptyset$ gives $B[0]=\tilde{B}\emptyset=\emptyset$. So $A\bullet A$ and $B\bullet A$ are finitary species with $\widetilde{A\bullet A}=\tilde{A}\circ\tilde{A}$ and $\widetilde{B\bullet A}=\tilde{B}\circ\tilde{A}$; the functor isomorphism and (J1) give $A\bullet A\cong B\bullet A$, hence equal cycle indices, and by (C)

$$Z_A\circ Z_A=Z_{A\bullet A}=Z_{B\bullet A}=Z_B\circ Z_A.$$

Here $H:=Z_A$ has zero constant term ($a_0=0$) and $a_1\geq 1$, so Lemma 5.17 cancels Z_A on the right: $Z_A=Z_B$. But B is flat, so $Z_B\in\mathbb{Q}[[p_1]]$; hence $Z_A\in\mathbb{Q}[[p_1]]$, i.e. A is flat and M is polynomial (F′). $\square$

This is precisely the cancellation of the trailing M that Proposition 5.10 forbade at the level of limit preservation — legitimate here because it operates on the *faithful* cycle-index invariant, which the single self-composite does determine. It reaches the monads no stabilizer count can: the free commutative magma (structure counts $1,1,3,15,105,\dots=(2n-3)!!$, $a_0=0$, a balanced 2^k-leaf tree with automorphism group $S_2\wr\cdots\wr S_2$, k factors) has unbounded nesting depth yet falls to Theorem 5.18 in one stroke.

5.8. **The unital corner: symmetry decides.** After Theorems 5.18 and 5.16 the residual is: $M=\tilde{A}$ analytic, $a_0=|M\emptyset|$ nonempty-finite, and unbounded wreath depth $\mathfrak{h}(A)=\infty$. Cancellation needs $a_0=0$; the depth invariant needs $\mathfrak{h} < \infty$; neither applies. This corner is inhabited, and — the surprise of this section — it contains monads on *both* sides of the converse.

Its canonical inhabitant is the *free commutative unital magma U*: the free-algebra monad of one commutative binary operation μ and a nullary unit e absorbed by μ ($\mu(x, e) = x$). Its species has $U[0] = \{e\}$ ($a_0 = 1$) and $U[n] = \{\text{commutative binary trees on } n \text{ leaves}\}$, $|U[n]| = (2n-3)!!$; the balanced 2^k-leaf tree has stabilizer $S_2 \wr \cdots \wr S_2$ (k factors), so $\mathfrak{h}(U) = \infty$ while every component is finite and $a_0 = 1$. It was the sharpest open case of the earlier form of this theorem. We show necessity *holds* not only at U and its fully symmetric family, but for every unital operad with a commutative binary composite and $\Delta_{C_2} \notin \mathcal{S}$ — an exact, operad-theoretic condition strictly weaker than full symmetry — and then that it *fails* one operad over, for a single ternary operation with cyclic symmetry.

What makes necessity hold: an escape witness and an exact criterion. The escape from the polynomial constraint of Proposition 5.12 is realised by a single, robust witness, built from any commutative binary composite and the unit; whether it escapes is then decided by an exact operad-theoretic condition, strictly finer than the commutativity of that composite.

Recall $\mathcal{S} := \{[\mathrm{Aut}(t)] : t \text{ a single } O\text{-tree, each label once}\}$ (conjugacy classes of automorphism groups of single trees) and, from Proposition 5.12, that every point-stabilizer of $(\llbracket r \rrbracket \circ M)[m]$ is a *block-fixing product* of members of $\mathcal{S}$ on pairwise disjoint supports; write $\mathcal{Y}$ for the set of such products. Put $\Delta_{C_2} := \langle (0\,1)(2\,3) \rangle$ (order 2, cycle type $(2,2)$). It is *support-indecomposable*: its two orbits $\{0,1\}, \{2,3\}$ cannot be split off as a direct product (the restrictions generate $S_2 \times S_2$ of order $4 \neq 2$). Hence:

Lemma 5.19 (Escape test)**.** *If $(M \circ M)[4]$ realises a point-stabilizer conjugate to Δ_{C_2} and $\Delta_{C_2} \notin \mathcal{S}$, then $M \circ M \not\cong \llbracket r \rrbracket \circ M$ for every polynomial $\llbracket r \rrbracket$.*

Proof. Were Δ_{C_2} a stabilizer of some $(\llbracket r \rrbracket \circ M)[4]$, it would lie in $\mathcal{Y}$; by support-indecomposability it could not factor across the two orbits, forcing a single tree t with $\mathrm{Aut}(t) = \Delta_{C_2}$, i.e. $\Delta_{C_2} \in \mathcal{S}$. So under $\Delta_{C_2} \notin \mathcal{S}$ the class $[\Delta_{C_2}]$ occurs in the stabilizer multiset of $(M \circ M)[4]$ but in that of no $(\llbracket r \rrbracket \circ M)[4]$; Lemma 5.11 gives the non-isomorphism. □

Lemma 5.20 (The correlated-swap witness)**.** *Let μ be any* commutative *binary composite of O ($\mu(x, y) = \mu(y, x)$) and suppose $a_0 \geq 1$. With V_i the inner leaf on label i and E' the inner unit e placed as an* outer *leaf — a rigid, label-free peg — set*

$$W_L = \mu(V_0, \mu(V_2, E')), \quad W_R = \mu(V_1, \mu(V_3, E')), \quad w = \mu(W_L, W_R) \in (M \circ M)[4].$$

Then $\mathrm{Aut}(w) = \langle (0\,1)(2\,3) \rangle = \Delta_{C_2}$.

Proof. W_L is rigid: the children of its outer μ are V_0 (a labelled leaf) and $\mu(V_2, E')$ (a two-node tree), which are non-isomorphic, so μ's slot-symmetry cannot exchange them (commutativity is irrelevant for non-isomorphic children); and inside $\mu(V_2, E')$ the children V_2 and E' are non-isomorphic. So $\mathrm{Aut}(W_L) = 1$, likewise $\mathrm{Aut}(W_R) = 1$, and $W_L \cong W_R$ via $0 \mapsto 1$, $2 \mapsto 3$. In $w = \mu(W_L, W_R)$ the two children are isomorphic, so commutativity of μ supplies the swap $W_L \leftrightarrow W_R$, realised on labels as $(0\,1)(2\,3)$; fixing the children gives $\mathrm{Aut}(W_L) \times \mathrm{Aut}(W_R) = 1$, and no single transposition preserves w. Adjoining further operations to O creates no isomorphism among these specific μ/E'-trees. □

The lemma uses *only* a commutative binary composite and $a_0 \geq 1$; no full symmetry. Whether the witness *escapes* — $\Delta_{C_2} \notin \mathcal{S}$ — is decided by a dichotomy in O.

Proposition 5.21 (Characterisation of $\Delta_{C_2} \in \mathcal{S}$)**.** *For a unital operad O, Δ_{C_2} is the automorphism group of a single O-tree* (*all four labels moved*) *if and only if one of:*

- (M1) *O has* both *a rigid binary composite ν* (*a 2-label single tree with trivial automorphism group*) and *a commutative binary composite μ; or*
- (M2) *O has a derived operation whose slot-symmetry, on some 4 leaf-slots* (*all remaining slots filled by the unit*)*, is exactly the locked double-swap $\langle(0\,1)(2\,3)\rangle$.*

Proof. ($\Leftarrow$) (M1): $t = \mu(\nu(V_0, V_2), \nu(V_1, V_3))$ has isomorphic rigid children $\nu(0, 2) \cong \nu(1, 3)$ swapped by μ, so $\mathrm{Aut}(t) = \langle(0\,1)(2\,3)\rangle$. (M2): the four-leaf tree $g(V_0, V_1, V_2, V_3)$ (remaining slots units) has $\mathrm{Aut}(t)$ equal to the derived symmetry $\langle(0\,1)(2\,3)\rangle$.

($\Rightarrow$) Let $\mathrm{Aut}(t) = \langle\tau\rangle$, $\tau = (0\,1)(2\,3)$, and put $a = \mathrm{lca}(0, 1)$, $b = \mathrm{lca}(2, 3)$; τ fixes a and b. *Key move:* if a node w is fixed by τ with $\tau(t_w) = t_w$ (t_w the subtree at w), then acting by $\tau|_{t_w}$ on t_w and by the identity on its siblings is an automorphism of t — applying an automorphism to one child-subtree while fixing its siblings is always legitimate and requires no transposition inside any slot-symmetry group. If a, b are incomparable, restricting τ to t_a gives an automorphism acting as $(0\,1)$ alone, contradicting $\mathrm{Aut}(t) = \langle\tau\rangle$; if $b < a$ strictly, $2, 3$ lie under one child D of a and restricting to t_D gives $(2\,3)$ alone — again a contradiction. Hence $a = b =: v$, at which τ swaps children isomorphically; the four labels distribute over v's children with a τ-symmetric count, and, τ being fixed-point-free, only $(1, 1, 1, 1)$ or $(2, 2)$ occur. In case $(1, 1, 1, 1)$ the four label-leaves are distinct children of v and every other child is a unit, so $\mathrm{Aut}(t)$ is the derived 4-slot symmetry $\langle(0\,1)(2\,3)\rangle$ — (M2). In case $(2, 2)$ two 2-label children A, B are swapped by τ; internal automorphisms of A, B extend to automorphisms of t, so $\mathrm{Aut}(t) = \langle\tau\rangle$ forces A, B rigid — a rigid binary composite — while the swap of A, B at v is a commutative binary composite — (M1). □

Theorem 5.22 (The escape criterion)**.** *Let $M = \tilde{A}$ be the free-algebra monad of a unital, non-flat operad O satisfying*

(i) *every binary composite is commutative* (*equivalently, O has no rigid 2-label single tree*), *and*
(ii) *no derived operation has locked-double-swap symmetry on any* 4 *leaf-slots* ($\neg$(M2)).

Then $M \circ M \not\cong [\![r]\!] \circ M$ for every polynomial $[\![r]\!]$; the essential image of $[\![-]\!]_M$ is not closed under composition, and — M being non-polynomial — M-containers compose $\iff$ M is polynomial for every such M.

Proof. By (i) and non-flatness some operation of arity ≥ 2 yields, on filling $n-2$ slots with the unit, a binary composite, commutative by (i); this is the μ of Lemma 5.20, and $a_0 \geq 1$ by unitality, so $\mathrm{Aut}(w) = \Delta_{C_2}$ occurs in $(M \circ M)[4]$. By (i) there is no rigid binary composite, so $\neg$(M1); by (ii), $\neg$(M2); Proposition 5.21 gives $\Delta_{C_2} \notin \mathcal{S}$. The Escape test (Lemma 5.19) concludes. $\square$

Full symmetry is the special case in which the two hypotheses are automatic.

Corollary 5.23 (Full symmetry gives necessity). *Let O be* fully symmetric unital*: every operation of arity ≥ 2 has full slot-symmetry $H_g = S_{\text{arity}}$, there is a nullary unit e — the* unique *nullary — absorbed by every operation, the only unary operation is the identity, and some operation has arity ≥ 2. Then its free-algebra monad $M = \tilde{A}$ satisfies the hypotheses of Theorem 5.22, so necessity holds. This includes the free commutative unital magma $U = U_2$* (*the corner inhabitant*) *and its n-ary analogues U_n for all $n \geq 2$.*

Proof. M is analytic, non-flat (some $H_g = S_{\geq 2} \neq 1$), with $a_0 = 1$ (the unique nullary) and $\mathfrak{h} = \infty$. Every S_k ($k \geq 2$) contains a transposition, so no binary composite is rigid: (i) holds. A derived symmetry on 4 leaf-slots is a symmetric group acting on the slot-orbit (full symmetry propagates through the tree), never the order-2 locked swap: (ii) holds. Apply Theorem 5.22. $\square$

The hypothesis that e is the *unique* nullary keeps the normal form clean: with every nullary absorbed, no leaf of a reduced tree is a unit, and the witness w is built solely from labelled leaves and the outer peg E'. (A second, non-absorbed nullary would sit as a rigid single-element peg, contributing only a trivial direct factor to every stabilizer; the argument survives, but we do not need that generality for the corollary.) Theorem 5.22 itself places no bound on $a_0 = |M\emptyset|$: the residual case $a_0 \geq 2$, $\mathfrak{h} = \infty$ is not empty and is covered whenever $\Delta_{C_2} \notin \mathcal{S}$, falling under the open converse (Problem 5.26) only when $\Delta_{C_2} \in \mathcal{S}$.

Theorem 5.22 is *strictly* stronger than its full-symmetry corollary. The operad of a single 4-ary operation with slot-symmetry the alternating group A_4, and the one with slot-symmetry the Klein four-group V_4, both satisfy (i)+(ii) — so necessity holds — yet neither is fully symmetric; the V_4-operad even has a rigid 3-label composite $g(x, y, z, e)$ (so it fails the "no rigid tree of any arity" property that the full-symmetry proof used) while still escaping, because a rigid *ternary* composite does not put Δ_{C_2} into $\mathcal{S}$. The correct axis is (i)+(ii) — a commutative binary composite together with $\Delta_{C_2} \notin \mathcal{S}$ — not full symmetry.

The mechanism is a *correlated block-swap*: the unit builds a rigid block W_L with no internal symmetry, and a commutative binary composite swaps two equal copies, giving $(0\,1)(2\,3)$ — a symmetry moving $0 \leftrightarrow 1$ *only together with* $2 \leftrightarrow 3$. A block-fixing product cannot correlate two blocks: it offers them independently ($S_2 \times S_2$, order 4) or fuses them into one symmetric tree (D_4, order 8); order 2 is unreachable unless a single tree already realises it, which (i)+(ii) forbid.

Partial symmetry: necessity fails. Full symmetry was load-bearing. Weaken it to *cyclic* slot-symmetry and the converse collapses.

Theorem 5.24 (The C_3 operad refutes necessity)**.** *Let O be the operad freely generated by one ternary operation ω with cyclic slot-symmetry $H_\omega = C_3 = \langle(0\,1\,2)\rangle$ (not S_3) and an absorbed nullary unit e, and let $M = \tilde{A}$ ($A = O$) be its free-algebra monad. Then M is analytic and non-polynomial (non-flat, $H_\omega = C_3 \neq 1$), with $a_0 = 1$ and $\mathfrak{h} = \infty$ — it inhabits the same residual as U. Nevertheless the essential image of $[\![-]\!]_M$* is *closed under composition: for every polynomial p and every polynomial q with at least one positive-arity shape,*

$$[\![p]\!]_M \circ [\![q]\!]_M \;\cong\; [\![p \triangleleft L]\!]_M, \qquad L = (\mathbb{N},\ n \mapsto [n]) \text{ the list (free-monoid) container}$$

(and a constant q, with only nullary shapes, gives a constant functor $[\![p]\!]_M \circ [\![q]\!]_M$, itself an M-container functor, so closure holds in that degenerate case too). Hence composition-closure does not *imply polynomiality; the necessity direction of Theorem 1.3 is false without a symmetry hypothesis.*

Lemma 5.25 (Only three identical)**.** *For the C_3 operad, the only source of a nontrivial label-automorphism in a structure of $[\![p]\!] \circ M \circ [\![q]\!] \circ M$ (p, q polynomial) is a ternary ω-node all three of whose children are isomorphic (contributing a C_3). The binary composite $\mu = \omega(x, y, e)$ is* non-commutative *(C_3 has no transposition), so two identical arguments are never swapped, and the flat p, q layers contribute only direct products. Consequently every point-stabilizer is an iterated C_3-wreath assembled by direct products, and each of its support-indecomposable factors is the automorphism group of a single O-tree.*

Proof idea. An automorphism of a rooted tree permutes each node's children within isomorphism classes using only that node's slot-symmetry. Under C_3 a 3-cycle fixes a configuration only when all three children coincide (every 3-cycle moves each slot to a non-identical one), and $\mu = \omega(-, -, e)$ has trivial residual symmetry since C_3 has no transposition fixing the unit slot; flat layers have distinguishable positions. So the only generator of label-symmetry is the three-identical C_3, the stabilizer is an iterated C_3-wreath built by direct products, and each support-indecomposable factor is realised by a single balanced O-tree. $\square$

Proof of Theorem 5.24. Two analytic-type functors are isomorphic iff they carry the same multiset of *molecule types* $(n, [G])$ (arity and stabilizer conjugacy class), with multiplicity, on every component — the coefficientwise Joyal reconstruction, valid for infinite components. By Lemma 5.25 the molecule types of $M \circ [\![q]\!] \circ M$ (any q with a positive-arity shape) are exactly the Young products of single-tree automorphism groups; the list composite $L \circ M$ realises the same set (a list of single trees and units has a block-fixing-product stabilizer), and every such type recurs with multiplicity $\aleph_0$ (pad by outer pegs, resp. appended units, without changing the stabilizer). Matching multisets gives $M \circ [\![q]\!] \circ M \cong L \circ M$; whiskering by $[\![p]\!]$,

$$[\![p]\!] \circ M \circ [\![q]\!] \circ M \;\cong\; [\![p]\!] \circ L \circ M = ([\![p]\!] \circ [\![L]\!]) \circ M \cong [\![p \triangleleft L]\!] \circ M = [\![p \triangleleft L]\!]_M.$$

The case $p = q = \mathsf{y}$ is $M \circ M \cong L \circ M$. As M is non-flat it is non-polynomial, so the closed image does not force polynomiality. □

The multiset match is *arity-preserving*, as an analytic-functor isomorphism demands. At each fixed arity n the components $(M \circ M)[n]$ and $(L \circ M)[n]$ are finite, and it is their molecule multisets *at* n, not merely aggregated over all n, that must agree. The bijection is explicit. A molecule of $M \circ M$ at arity n is a normal-form two-level O-tree on $[n]$: an outer O-tree whose leaves are inner O-trees, the inner label-sets partitioning $[n]$ — equivalently a unit-completed C_3-tree on n labels. A molecule of $L \circ M$ at arity n is a *list* of inner O-trees whose label-sets partition $[n]$. Sending a two-level tree to the list of its inner subtrees, read in the canonical order of the outer skeleton, is an arity-preserving bijection between unit-completed C_3-trees on $[n]$ and lists of C_3-trees on set-partitions of $\{1, \ldots, n\}$; by Lemma 5.25 it carries each stabilizer isomorphically (both are the same iterated-C_3 block-fixing product). So the multiplicities agree at every n — each type recurring $\aleph_0$ times across arities as outer pegs, resp. appended units, are adjoined — which is what the isomorphism of Theorem 5.24 requires.

The contrast with Corollary 5.23 is exact — the same construction seen from two operads. The correlated-swap witness w requires the outer μ to *swap* its two equal children, which needs μ commutative; under C_3, $\mu = \omega(-, -, e)$ is non-commutative, so w has trivial automorphism group and no escape occurs. (Verified: the molecule type-sets of $M \circ M$ and $L \circ M$ coincide for $m \leq 4$; the same escape search that finds Δ_{C_2} for the *fully symmetric* S_3-operad at $m = 4$ finds *nothing* for C_3.)

The dividing line. Theorem 5.22 and Theorem 5.24 sit on opposite sides of a single, exact condition on the operad — and it is *not* the commutativity of the binary unit-composite, which a first reading of the two endpoints might suggest. The escape witness w of Lemma 5.20 exists whenever O has a commutative binary composite; whether it *escapes* is governed by

$$\Delta_{C_2} \notin \mathcal{S} \qquad \big(\text{equivalently } \neg(\mathrm{M1}) \wedge \neg(\mathrm{M2}),\ \text{Proposition 5.21}\big),$$

which is strictly finer than any of the obvious candidates. A commutative binary composite is not enough: if O *also* has a rigid binary composite ν (call this operad $O_{\mu\nu}$) then $\mu(\nu(0,2),\nu(1,3))$ realises Δ_{C_2} as a single-tree automorphism group (M1), and the witness fails to escape. "No rigid binary composite" is not enough either: the pentagon operad D_5 has every binary composite commutative yet a derived operation $g(0,1,2,3,e)$ with automorphism group Δ_{C_2} (M2). And full symmetry is more than necessary, as the A_4- and V_4-operads show.

candidate dividing line	refuted by	mechanism
"has a commutative binary composite"	$O_{\mu\nu}$	rigid ν too $\Rightarrow \Delta_{C_2} \in \mathcal{S}$ (M1) $\Rightarrow$ composes
"no rigid binary composite"	pentagon D_5	$g(0,1,2,3,e)$ has $\mathrm{Aut} \cong \Delta_{C_2}$ (M2) $\Rightarrow$ composes
"no rigid tree of any arity"	V_4-operad	rigid *ternary* composite, yet $\Delta_{C_2} \notin \mathcal{S} \Rightarrow$ necessity
"fully symmetric"	A_4, V_4	not fully symmetric, yet $\Delta_{C_2} \notin \mathcal{S} \Rightarrow$ necessity

The invariant that survives all four candidates, *among operads that possess a commutative binary composite*, is $\Delta_{C_2} \in \mathcal{S}$, read off the operad by (M1)$\vee$(M2). The scope qualifier is essential, and the C_3 operad shows why. Its only binary composite $\omega(x,y,e)$ is rigid and non-commutative, so it has *no* commutative binary composite at all: the swap witness w of Lemma 5.20 cannot even be built, and the image closes (Theorem 5.24) for a reason orthogonal to Δ_{C_2}. Commutativity of a binary composite is thus a genuine *precondition* for the escape — it is what drives the block-swap — and only *given* such a composite does the finer condition $\Delta_{C_2} \notin \mathcal{S}$ decide the outcome. A fully symmetric unital operad is the clean extreme: it has a commutative binary composite and $\Delta_{C_2} \notin \mathcal{S}$ (Corollary 5.23).

Precisely, Theorem 5.22 establishes the forward direction: an operad with a commutative binary composite and $\Delta_{C_2} \notin \mathcal{S}$ — equivalently, satisfying (i)+(ii) — has $M \circ M \not\cong [\![r]\!] \circ M$. Necessity can fail in two disjoint ways: because $\Delta_{C_2} \in \mathcal{S}$ (a rigid or locked-swap companion, cases M1/M2), or because no commutative binary composite exists at all (as for C_3). The converse of Theorem 5.22, within the class that *does* have a commutative binary composite, is the natural completion.

Problem 5.26 (The composition biconditional, among operads with a commutative binary composite)**.** Restrict to unital, non-flat operads O possessing a commutative binary composite. Is the essential image of $[\![-]\!]_M$ closed under composition *if and only if* $\Delta_{C_2} \in \mathcal{S}$ (equivalently (M1)$\vee$(M2))? Theorem 5.22 proves the "only if" (contrapositive: $\Delta_{C_2} \notin \mathcal{S} \Rightarrow$ not closed). For the "if" direction the mechanism of Theorem 5.24 — matching molecule multisets to obtain $M \circ M \cong [\![r]\!] \circ M$ — becomes available once $\Delta_{C_2} \in \mathcal{S}$ supplies the missing single tree, and the all-binary inhabitant $O_{\mu\nu}$ of case (M1) agrees with it in computation for $m \leq 4$; a proof for all m and all such O remains open. Operads with *no* commutative binary composite (such as the C_3 operad) lie outside this dichotomy and may close their image for the separate reason that no swap witness exists.

Assembling sufficiency with the conditional converse:

Theorem 5.27 (Composition and polynomiality; final form)**.** Sufficiency (unconditional): *if M is polynomial then the essential image of $[\![-]\!]_M$ is closed under composition, with $p \lhd_M q = p \lhd \lceil M \rceil \lhd q$ (Proposition 5.2).* Necessity (conditional): *composition-closure forces M polynomial for M super-polynomial (Proposition 5.6), for analytic M with $M\emptyset = \emptyset$ (Theorem 5.18), for analytic M of bounded wreath depth (Theorem 5.16), and for the free-algebra monad of every unital non-flat operad with a commutative binary composite and $\Delta_{C_2} \notin \mathcal{S}$ (Theorem 5.22) — a class containing every fully symmetric unital operad (Corollary 5.23, hence the corner case U and the U_n) and strictly more (the A_4- and V_4-operads); it* fails *for the C_3 operad (Theorem 5.24). The dividing line, among operads that possess a commutative binary composite, is $\Delta_{C_2} \in \mathcal{S}$, read off the operad by (M1)$\vee$(M2) (Proposition 5.21). Consequently* within the fully symmetric world — which contains every applied effect monad: multiset, distribution, powerset, symmetric powers — composition-closure $\iff$ flatness of the species $\iff$ M polynomial. *The earlier reliance on an unproved species-theoretic* Plethysm Lemma *is removed (Theorem 5.18), the formerly open corner U is resolved (necessity holds), and the necessity criterion is sharpened from full symmetry to the exact condition $\Delta_{C_2} \notin \mathcal{S}$; the C_3 operad shows the biconditional cannot be extended to all unital analytic monads.*

5.9. **Independence.**

Proof of Corollary 1.4. The writer $E\times(-)$ is polynomial ($E\times(-) = \sum_{e\in E}(-)^1$), so it composes by Proposition 5.2; it is not codense ($\Phi(1) = |E|^{|E|} \neq |E|$ for $|E| \geq 2$), so it is not full by Theorem 1.2. The distribution monad D is affine and codense (Example 4.3), so it is full by Theorem 1.2; it is not polynomial (it does not preserve wide pullbacks — joints are not determined by marginals), so it fails composition by Corollary 5.7. Codensity and polynomiality are therefore logically independent properties of M. ☐

6. Conclusion

For a monad M on **Set**, the effectful container extension $[\![-]\!]_M = [\![-]\!] \circ M$ names endofunctors faithfully always, names them *fully* exactly when M is codense, and has an image closed under composition when M is polynomial — and, within the fully symmetric world that holds the applied effect monads, *only* when M is polynomial. These are two different Kan-theoretic invariants of the effect monad, and across the standard effects they pull in opposite directions: probability is faithful but not composable, error is composable but not faithful. The folklore "composes iff commutative and affine" is simply wrong — D is affine and commutative but does not compose, Maybe is neither yet does — and the true obstruction to composition is a representability question (flatness of a species), not a cohomology class or a distributive-law existence question. That representability governs composition turns out to require a symmetry hypothesis: the C_3 operad, whose non-polynomial monad nonetheless composes, marks the exact boundary.

For the compositional modelling of effectful agents this draws a sharp line. An agent that "may decline" (Maybe-positions) chains cleanly into other agents but cannot always be recovered from its behaviour; an agent that "answers probabilistically" (D-positions) is recoverable from its behaviour but does not chain into a single probabilistic agent. One pays for declining in faithfulness and for probability in composability, and the invariant that sets each price is now explicit.

Three questions remain, each precisely located.

(1) **The composition converse, and its boundary (Problem 5.26).** Does closure of the essential image under $\circ$ force M to be polynomial? *Conditionally yes, and in general no.* The converse holds for super-polynomial monads (the cardinality law), for analytic monads with $M\emptyset = \emptyset$ (plethysm right-cancellation, Lemma 5.17, at any wreath depth), for analytic monads of bounded wreath depth (the imprimitivity-depth invariant h, Definition 5.14), and for the free-algebra monad of every unital non-flat operad with a commutative binary composite and $\Delta_{C_2} \notin \mathcal{S}$ (a correlated block-swap, Theorem 5.22) — a class strictly larger than the fully symmetric operads, resolving the formerly open corner U in favour of necessity. But the converse *fails* for the C_3 operad (Theorem 5.24): a non-polynomial monad whose image is nonetheless composition-closed, with $M \circ M \cong L \circ M$ for the list monad L. The trailing-M obstruction, sharpened to the "no information" statement of Proposition 5.10 at the level of limit preservation, is circumvented on the finer stabilizer and cycle-index invariants; and, *given* a commutative binary composite, the exact dividing line is the combinatorial condition $\Delta_{C_2} \in \mathcal{S}$ — whether the locked double-swap is a single-tree automorphism group, read off by (M1)$\vee$(M2) (Proposition 5.21) — *not* the mere commutativity itself, which is only a precondition. What remains open is the converse of Theorem 5.22, within the class possessing a commutative binary composite: does $\Delta_{C_2} \in \mathcal{S}$ force the image closed, completing the biconditional (Problem 5.26)?

(2) **Rigidity of D (Lemma 4.4).** Full faithfulness of $[\![[]\!]_M\mathsf{D}]-$ is conditional on $\mathrm{Nat}(\mathsf{D},\mathsf{D}) = \{\mathrm{id}\}$; the reduction of Example 4.3 is unconditional. A clean proof of this rigidity fact would remove the one assumption in Theorem 1.2's probabilistic case.

(3) **Kleisli-bifunctoriality (Remark 5.5).** The composition product is defined on objects and is strong semigroupal on pure morphisms; it upgrades to the full Kleisli category exactly when M is codense. Thus the two theorems meet: composition needs polynomiality to

exist and codensity to become bifunctorial on effectful morphisms. The monads for which both hold — the affine polynomial monads — are worth isolating.

References

[1] D. Ahman and T. Uustalu, *Directed containers as categories*, Electron. Proc. Theor. Comput. Sci. **207** (2016), 89–98. arXiv:1604.01187.

[2] F. Bergeron, G. Labelle, and P. Leroux, *Combinatorial Species and Tree-like Structures*, Encyclopedia of Mathematics and its Applications **67**, Cambridge University Press, 1998.

[3] A. Dorta, C. B. Jarvis, and N. Niu, *Monoidal structures on generalized polynomial categories*, arXiv:2305.05655.

[4] N. Gambino and J. Kock, *Polynomial functors and polynomial monads*, Math. Proc. Cambridge Philos. Soc. **154** (2013), no. 1, 153–192. arXiv:0906.4931.

[5] A. Joyal, *Foncteurs analytiques et espèces de structures*, in *Combinatoire énumérative* (Montréal, 1985), Lecture Notes in Math. **1234**, Springer, 1986, 126–159.

[6] N. Niu and D. I. Spivak, *Polynomial Functors: A Mathematical Theory of Interaction*, book draft, 2023. arXiv:2312.00990.

Kodamai